

Exercise 1. General intro and methods

Task set A:

Multiple choice

1. Which of the following sequences is palindromic?

5' GCAACG 3'

5' GCAT 3'

5' GCATGC 3' (GCATGC -> CGTACG)

5' GCAACGC 3'

5' GGAAAAGG 3'

2. Which of the following is **not** a good feature of a plasmid vector?

Multiple cloning site

Antibiotic resistance gene

Origin of replication

RNA polymerase encoding gene

Blue/white screening system

Because:

A good plasmid vector typically has:

- **Origin of replication (Ori):** ensures replication inside host cells.
- **Multiple cloning site (MCS):** region with several restriction sites for easy insertion of foreign DNA.
- **Antibiotic resistance gene:** selectable marker for identifying transformed cells.
- **Blue/white screening system:** useful reporter system to differentiate recombinant from non-recombinant plasmids.

But a plasmid vector **does not need to encode RNA polymerase**, since the host cell already provides the necessary RNA polymerases. Including such a gene would make the plasmid unnecessarily large and inefficient.

3. Which of the following molecular components would be present in a DNA sequencing reaction by the chain termination method, but *not* in a PCR reaction to amplify a particular gene?

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At least one DNA primer

Dideoxynucleotides

A DNA polymerase

Template DNA

Deoxynucleotides

Because:

- **PCR reaction** requires:
 - Template DNA
 - A DNA polymerase (usually Taq)
 - At least one DNA primer (usually two: forward and reverse)
 - Deoxynucleotides (dNTPs)
- **Sanger DNA sequencing (chain termination method)** requires all of the above **plus**:
 - **Dideoxynucleotides (ddNTPs)** → these lack a 3'-OH group, so when incorporated, they terminate chain elongation.

☞ Therefore, **ddNTPs are unique to sequencing reactions** and are not used in standard PCR amplification.

4. The technique of _____ is used to transfer DNA from an agarose gel to a membrane.

southern blotting

qPCR

restriction mapping

western blotting

northern blotting

Because:

- **Southern blotting** → transfers **DNA** from an agarose gel to a membrane, followed by hybridization with a labeled probe.
- **Northern blotting** → similar method, but for **RNA**.
- **Western blotting** → used for **proteins**.
- **qPCR** → quantifies DNA amplification in real-time, not a transfer technique.
- **Restriction mapping** → method to determine restriction enzyme cut sites on DNA, not a transfer technique.

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☞ So, transferring DNA from agarose gel to a membrane = **Southern blotting**.

5. In terms of function, the “opposite” of a polymerase is a

nuclease.

ligase.

phosphatase.

β -galactosidase.

luciferase.

Because:

- **Polymerase** → *builds* polymers of nucleic acids by adding nucleotides (DNA or RNA synthesis).
- **Nuclease** → *breaks down* nucleic acids by cleaving phosphodiester bonds (the opposite of building).
- **Ligase** → joins fragments of DNA, not the opposite of polymerase.
- **Phosphatase** → removes phosphate groups, unrelated to nucleotide chain synthesis.
- **β -galactosidase** → enzyme that cleaves lactose, unrelated.
- **Luciferase** → produces light in bioluminescent reactions, unrelated.

☞ Therefore, the opposite of a polymerase (builder) is a **nuclease (breaker)**.

6. Which statement regarding PCR is FALSE?

PCR permits the exponential amplification of a desired DNA sequence by using primers that anneal to the sequence of interest

qPCR is a method that allows quantification of the template

PCR depends on thermostable DNA polymerases

Deoxyribonucleotides are substrates in all PCR types

Reverse transcriptase-PCR (RT-PCR) uses RNA polymerase to convert DNA to RNA for use in a polymerase chain reaction.

Because:

- **PCR permits exponential amplification** → TRUE
- **qPCR allows quantification** → TRUE
- **PCR depends on thermostable DNA polymerases (e.g., Taq)** → TRUE

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- **dNTPs are substrates in PCR → TRUE**
- **RT-PCR → actually uses reverse transcriptase to convert RNA → cDNA (complementary DNA), which is then amplified by PCR.**
 - The statement says **RNA polymerase converts DNA → RNA**, which is incorrect.

☞ Therefore, that last statement is the **false one**.

7. Western blots detects _____, Northern blots detects _____, whereas Southern blot detects _____.

protein, DNA, RNA

protein, RNA, DNA

DNA, protein, RNA

RNA, DNA, protein

RNA, protein, DNA

Because:

Quick mnemonic:

- **SNoW DRoP → Southern = DNA, Northern = RNA, Western = Protein**

Task set B

Task set B contains 2 essay questions (4 in the final exam). The answer should be written using the space allocated in the box below each question.

1. Primer design. Consider the following DNA sequence. It is part of the *Escherichia coli* cloning vector pBR322. The part in red is the coding strand of the open reading frame of the tetracycline resistance gene (*tetR*) including the stop codon. You want to amplify the open reading frame by PCR

TTCTCATGCTTTGACAGCTTATCATCGATAAGCTTTAATGCGGTAGTTTATCACAGTTAAATTGCTAACGCAGTCAGGCAC
CGTGT**ATGAAATCTAACAATGCGCTCATCGTC**TCCTCGGCACCGTCACCCTGGATGCTGTAGGCATAGGCTTGGTTATG
CCGGTACTGCCGGGCCCTCTCGGGATATCGTCCATTCCGACAGCATCGCCAGTCACTATGGCGTGCTGCTAGCGCTATA
TGCGTTGATGCAATTTCTATGCGCACCCGTTCTCGGAGCACTGTCCGACCGCTTTGGCCGCCGCCAGTCCTGCTCGCTT
CGCTACTTGGAGCCACTATCGACTACGCGATCATGGCGACCACACCCGTCCTGTGGATCCTCTACGCCGGACGCATCGTG
GCCGGCATCACCGGCCGCCACAGGTGCGGTTGCTGGCGCCTATATCGCCGACATCACCGATGGGGAAGATCGGGCTCGCCA
CTTCGGGCTCATGAGCGCTTGTTTCGGCGTGGGTATGGTGGCAGGCCCGTGGCCGGGGGACTGTTGGGCGCCATCTCCT
TGCATGCACCATTCCTTGCGGCGGCGGTGCTCAACGGCCTCAACCTACTACTGGGCTGCTTCCTAATGCAGGAGTCGCAT
AAGGGAGAGCGTCGACCGATGCCCTTGAGAGCCTTCAACCCAGTCAGCTCCTTCCGGTGGGCGCGGGGCATGACTATCGT
CGCCGCACTTATGACTGTCTTCTTTATCATGCAACTCGTAGGACAGGTGCCGGCAGCGCTCTGGGTCAATTTTCGGCGAGG
ACCGCTTTCGCTGGAGCGCGACGATGATCGGCCTGTCGCTTGCAGTATTCGGAATCTTGACGCCCTCGCTCAAGCCTTC

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GTCACTGGTCCCGCCACCAAACGTTTCGGCGAGAAGCAGGCCATTATCGCCGGCATGGCGGCCGACGCGCTGGGCTACGT
CTTGCTGGCGTTTCGCGACGCGAGGCTGGATGGCCTTCCCCATTATGATTCTTCTCGCTTCCGGCGGCATCGGGATGCCCG
CGTTGCAGGCCATGCTGTCCAGGCAGGTAGATGACGACCATCAGGGACAGCTTCAAGGATCGCTCGCGGCTCTTACCAGC
CTAACTTCGATCACTGGACCGCTGATCGTCACGGCGATTTATGCCGCCTCGGCGAGCACATGGAACGGCTTCCGCTCGAT
TGTAGGCGCCGCCCTATACCTTGTCTGCCTCCCCGCGTTGCGTCGCGGTTCATGGAGCCGGGCCACCTCGACCTGAATGG
AAGCCGGCGGCACCTCGCTAACGGATTCACTCACTCCAAGAATTGGAGCCAATCAATTCTTGGGAGAACCTGCTGATTCGCGC
AAACCAACCCTTGGCAGAACATATCCATCGCGTCCGCCATCTCCAGCAGCCGCACGCGGCGCATCTCGGGCAGCGTTGGG
TCCTGGCCACGGGTGCGCATGATCGTGCTCCTGT

Design the primers (20 nucleotides long) that can be used to amplify the *tetR* open reading frame.
Give the sequences of the two required primers in the 5'-3' direction.

ANSWER:

Forward: 5'-ATGAAATCTAACAATGCGCT-3' (THE COMPLEMENT TO THE BEGINNING OF THE OPEN READING FRAME (ORF))

Reverse: 5'-TCAGGTCGAGGTGCCCCGGC-3' (THE REVERSE COMPLEMENT TO THE GREEN AREA - THE END OF THE OPEN READING FRAME (ORF))

Start and stop codons:

☒ Start Codons (Initiation Codons)

Primary (most common in bacteria, eukaryotes, and organelles):

- **AUG** → codes for **Methionine** (Met) in eukaryotes and bacteria (fMet in bacteria)

Alternative start codons (mostly in bacteria and mitochondria):

These normally code for other amino acids internally, but when used as the first codon, they are interpreted as Methionine (or fMet).

- **GUG**
- **UUG**
- **AUU**
- **AUA**
- **CUG**

Important notes:

- **AUG** is by far the dominant start codon.
- **GUG** and **UUG** are well-documented alternative start codons in *E. coli*.
- **CUG**, **AUA**, and **AUU** start codons occur almost exclusively in mitochondrial genomes or rare bacterial cases.

☐ Stop Codons (Termination Codons)

These do **not** encode amino acids; instead, they recruit release factors to terminate translation.

- **UAA** ("ochre")
- **UAG** ("amber")
- **UGA** ("opal")

All three are universal stop codons, although mitochondria sometimes reassigned UGA to code for tryptophan.

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2. Set up the individual steps in a RT-PCR experiment where you want to verify that a specific gene is expressed and which is the relative concentration. Include enzymes and substrates.

ANSWER:

in summary, an RT-PCR experiment to verify **expression** and **relative concentration** of a specific gene consists of:

1. **Isolating RNA,**
2. **Reverse transcribing RNA to cDNA** using reverse transcriptase and dNTPs,
3. **Amplifying the gene's cDNA by qPCR** with DNA polymerase, gene-specific primers, dNTPs, and a fluorescent reporter,
4. **Interpreting Ct values (normalized to a housekeeping gene)** to determine whether the gene is expressed and at what relative level.

The longer explanation:

1. Isolation of total RNA

Purpose: Obtain the RNA pool that may contain the mRNA of your gene of interest.

- Starting material (substrate): Cells or tissue containing your gene of interest.
- Key reagents/enzymes:
 - Lysis buffer with chaotropic salts (e.g. guanidinium thiocyanate)
 - RNase inhibitors (protect RNA)
 - Optional DNase I (enzyme to degrade contaminating genomic DNA; substrate = DNA)
- Product: Purified total RNA (mRNA + rRNA + tRNA), free of DNA.

This step ensures that any later signal comes from RNA (i.e. expression) and not from contaminating DNA.

2. Reverse transcription (RT): RNA → cDNA

Purpose: Convert mRNA of the specific gene into complementary DNA (cDNA), because PCR works on DNA, not RNA.

- Enzyme: Reverse transcriptase (e.g. M-MLV, SuperScript).
- Substrates:
 - RNA template (the total RNA, including the mRNA of your gene)
 - Primers (one of the following):
 - Oligo(dT) primers (anneal to poly(A) tails of mRNAs)
 - Random primers (bind throughout RNA population)
 - Gene-specific primer (binds only your gene's mRNA)
 - dNTPs: dATP, dCTP, dGTP, dTTP (building blocks for DNA synthesis)
- Buffer and cofactors: RT buffer with Mg^{2+} , RNase inhibitor.
- Product: cDNA mixture (including cDNA corresponding to your gene if it is expressed).

If your gene is not expressed, its specific cDNA will not be present in significant amount.

3. Quantitative PCR (qPCR) on cDNA: amplify and measure

Purpose: Specifically amplify the cDNA of your gene and measure how much is there (relative concentration).

- Enzyme: Thermostable DNA polymerase (e.g. Taq or hot-start Taq).
- Substrates and key components:
 - cDNA template from step 2 (contains your gene's cDNA if expressed)
 - Gene-specific forward and reverse primers (bind only your gene's cDNA)
 - dNTPs (dATP, dCTP, dGTP, dTTP)
 - PCR buffer with Mg^{2+}
 - Fluorescent system to monitor amplification:
 - SYBR Green (binds double-stranded DNA) *or*
 - TaqMan probe (sequence-specific fluorescent probe; substrate for polymerase's 5'→3' exonuclease activity)
- Control reactions:
 - Housekeeping/reference gene (e.g. GAPDH, ACTB) with its own primer pair
 - No-template control (NTC)
 - No-RT control (to check for DNA contamination)

During cycling (denaturation–annealing–extension), the DNA polymerase uses cDNA + primers + dNTPs as substrates to synthesize new DNA, and fluorescence increases proportional to the amount of amplified product.

4. Data analysis: Is the gene expressed? What is the relative concentration?

Purpose: Use qPCR output to decide if the gene is expressed and how much (relative to a reference).

- Readout: Ct (threshold cycle) values from qPCR.
 - If your gene's Ct is detectable (and specific), the gene is expressed.
 - If no Ct (or only in no-RT control), the gene is not detectably expressed.
- Relative concentration (relative expression):
 1. For each sample, compute
$$\Delta Ct = Ct_{\text{target gene}} - Ct_{\text{housekeeping gene}}$$
 2. Choose a reference condition (e.g. untreated cells) and compute
$$\Delta\Delta Ct = \Delta Ct_{\text{sample}} - \Delta Ct_{\text{reference}}$$
 3. Relative expression (fold change):
$$\text{Fold change} = 2^{-\Delta\Delta Ct}$$

This gives you a relative concentration of your gene's transcript between conditions, normalized to a housekeeping gene and thus independent of differences in RNA input or RT efficiency.

Esperanza's answer:

1. Harvest the cells.

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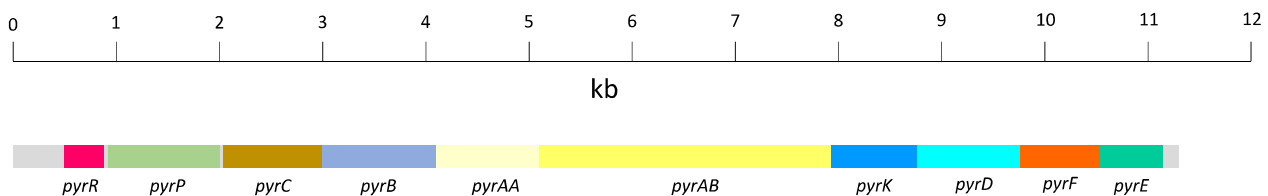
2. Extract RNA
3. Produce cDNA by a reverse transcriptase
4. Use specific primers in a PCR reaction with dNTPs, a heat stable DNA polymerase
5. Run a gel to verify the presence of a PCR product with the expected length
5. Monitor fluorescence development and determine the threshold cycle number (CT) from the two samples

Task set C.

Task set C contains two problem based questions (1 or 2 that will be longer in the final exam). The answer should be written using the space allocated in the box below each question.

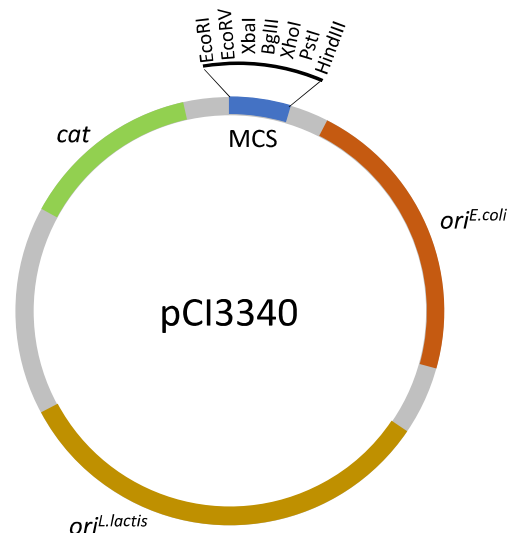
Problem 1. Cloning of a gene by PCR.

The *de novo* biosynthesis of the pyrimidine nucleotide UMP is catalysed by 6 enzymatic reactions. In *Bacillus subtilis*, the enzymes are encoded by the *pyr* operon. The genetic map is shown below:



In another Gram-positive bacterium, *Lactococcus lactis*, a mutant with a defective *pyrK* gene has been isolated. The mutant is unable to grow unless it is supplied with pyrimidines in the growth medium. To investigate whether the corresponding gene from *B. subtilis* can abolish the effect of the mutation by complementation, the *pyrK* gene from *B. subtilis* can be cloned in a plasmid vector, that subsequently will be established in *L. lactis*. As plasmid constructions in *L. lactis* is not straightforward, it was decided to use *Escherichia coli* as a construction host. Therefore, the shuttle vector pCI3340 (right) was chosen as carrier of the *pyrK* gene from *B. subtilis*, as it harbours two replication origins active in either *E. coli* or *L. lactis*.

Moreover, it has a chloramphenicol resistance gene that is active in both organisms. For cloning a PCR strategy was chosen. Below is the sequence of the *pyrK* region of the *B. subtilis* *pyr* operon shown together with a genetic map of pCI3340.



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A. Briefly describe the different steps in the cloning strategy. Details about the primers and the selected restriction enzymes to be used are subject to the next question.

ANSWER:

1. Harvest the cells and extract the DNA
2. Use specific primers in a PCR reaction with dNTPs, a heat stable DNA polymerase
3. Digest both the PCR product and plasmid pCI3340 with
4. Mix and add ligase
5. Transform to *E. coli* and select for Chloramphenicol resistance
6. Isolate plasmid DNA and investigate by gel electrophoresis and Sequencing
7. Transform to *L. lactis*

Explanation:

The cloning strategy begins with isolating genomic DNA from *Bacillus subtilis*, because this provides the template containing the *pyrK* gene. Once the DNA is obtained, PCR is performed using primers that specifically anneal to the *pyrK* sequence. PCR serves two essential functions: it selectively amplifies only the *pyrK* gene from the entire genome, and it allows engineered additions—such as restriction sites—to be included at the ends of the amplified product through primer design.

After amplification, both the PCR product and the plasmid vector pCI3340 are digested using chosen restriction enzymes. This ensures that the ends of the insert and vector are compatible for ligation. Using two different restriction enzymes generates non-identical overhangs, which forces the gene into the plasmid in the correct orientation and reduces unwanted self-ligations of the vector.

Next, DNA ligase is added to covalently join the insert and vector, creating a recombinant plasmid. This plasmid is then transformed into *E. coli*, where chloramphenicol selection ensures that only bacteria containing the plasmid survive. Because cloning is probabilistic and can produce incorrect constructs, plasmid DNA is isolated from *E. coli* and verified by gel electrophoresis and DNA sequencing to confirm the presence, orientation, and integrity of the *pyrK* gene. Once the correct construct is identified, it is finally transformed into *Lactococcus lactis* to test whether the cloned *pyrK* can complement the defective endogenous gene. Each step in this strategy reflects a fundamental principle of molecular cloning: isolate, amplify, engineer, assemble, verify, and ultimately express the gene in the target organism.

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Restriction endonucleases and their recognition sequences	
EcoRI	GAATTC
EcoRV	GATATC
XbaI	TCTAGA
BglII	AGATCT
XhoI	CTCGAG
PstI	CTGCAG
HindIII	AAGCTT

B. Design primers for PCR. Include suitable restriction enzyme sites in the primers. Use the table above and the map of pCI3340 to find the appropriate restriction enzymes and their sequences. The primers must be written in the 5' to 3' direction. Give the name of chosen restriction enzyme(s) after the primer sequence, and mark the restriction enzyme recognition sequence in the primer by underscoring. The primers should not exceed 30 bases.

ANSWER:

Forward: 5'- TCTAGAATGAAAAAAGCGTATTTGACGTAT-3' (XbaI)

Reverse: 5'- CTGCAGTTACAGCGCCACCTCCTGAGCTTT -3' (PstI)

Explanation:

The designed primers incorporate an XbaI site in the forward primer and a PstI site in the reverse primer. These restriction sites are intentionally added at the 5' ends of the primers so that the resulting PCR product contains specific restriction enzyme recognition sequences. Including these sites makes it possible to digest the PCR product and ligate it into the vector at matching sites.

The use of two different restriction enzymes is especially important because it enforces directional cloning: the gene can only be inserted in one orientation. This prevents issues such as reversed gene insertion, which would interfere with proper transcription and translation. Furthermore, restriction enzymes require a few extra nucleotides upstream of their recognition site to cut efficiently, which is why the restriction sites must be built into the primer sequence rather than relying on natural sites within the gene. Overall, the inclusion of restriction sites in the primers is a deliberate strategy to ensure precise and functional integration of the gene into the plasmid.

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C. Why is EcoRV restriction sites not a good choice in this experiment?

The EcoRV will cut in the pyrK gene.

Explanation:

EcoRV is not suitable for this cloning experiment because its recognition sequence (GATATC) occurs within the pyrK gene itself. If EcoRV were used to digest the PCR product or genomic DNA, it would cut inside the coding sequence and disrupt the gene. This would prevent the production of a full-length pyrK protein, making the cloning attempt unsuccessful from the start. A core principle in cloning is that restriction enzymes used for vector assembly must not cleave the gene being cloned unless this is specifically intended for engineering. Because the goal here is to preserve an intact functional gene for complementation, any enzyme that cuts within pyrK is immediately disqualified.

The construction was successfully established in *L. lactis* but it did not complement the defective *pyrK* gene. Total RNA from the transformant was isolated and used in a Reverse-Transcriptase PCR (RT-PCR) experiment, but no product could be identified using primers specific for the *B. subtilis pyrK* gene.

D. What could be the reason?

ANSWER:

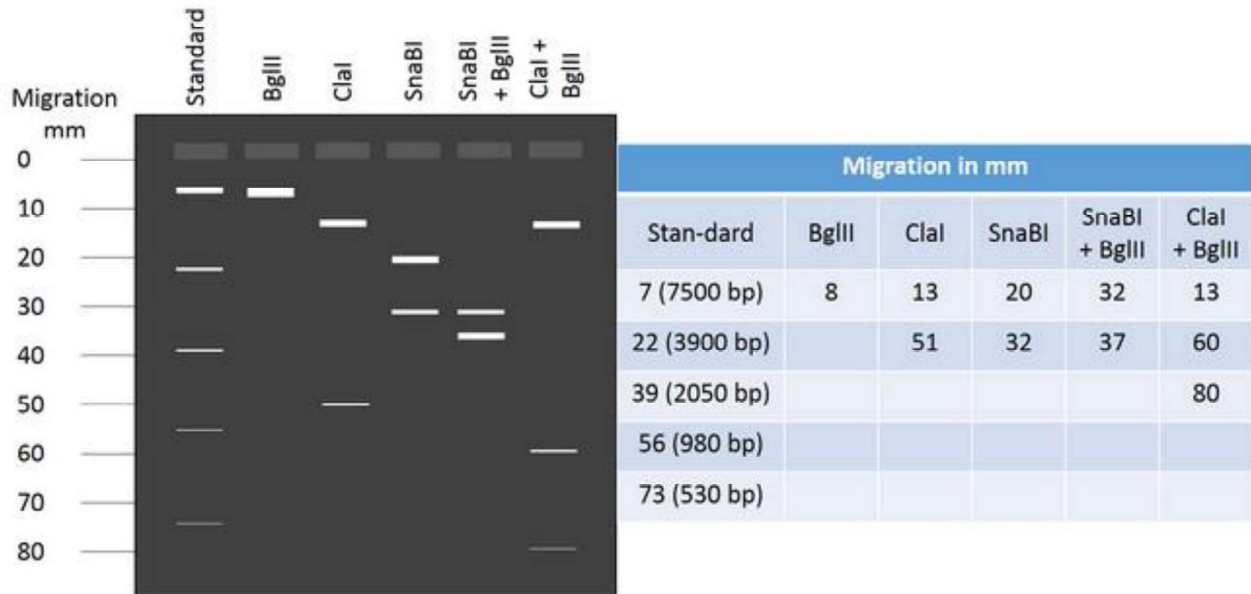
A promotor is missing in front of the pyrK gene in the plasmid

Explanation:

The failure to detect pyrK mRNA in *L. lactis* during the RT-PCR experiment is explained by the absence of a promoter upstream of the cloned gene. The PCR product contains only the pyrK coding region, and the chosen vector pCI3340 does not provide a promoter at the insertion site. As a result, even though the gene is physically present in *L. lactis*, there is no promoter recognized by the host's RNA polymerase to initiate transcription. Without transcription, mRNA is not produced, which in turn means no protein is translated. This explains both the negative RT-PCR result and the failure of the cloned gene to complement the pyrK defect. The underlying concept is that successful gene expression depends not only on the presence of the coding sequence but also on the regulatory elements required for transcription initiation.

Problem 2. Determination of DNA-fragments by electrophoresis.

The plasmid pRUT145 was extracted from the photosynthetic bacterium *Anabaena variabilis*. The plasmid DNA was digested with the Restriction enzymes *Sna*BI, *Cla*I, *Bgl*II and i combinations of the three enzymes. The digests were applied to an agarose gel together with a known size marker. An electronic image of the gel, and the corresponding migration distances are shown below. For the standard, the known fragment sizes are presented.



A. How many times do the individual restriction enzymes cut pRUT145?

ANSWERS

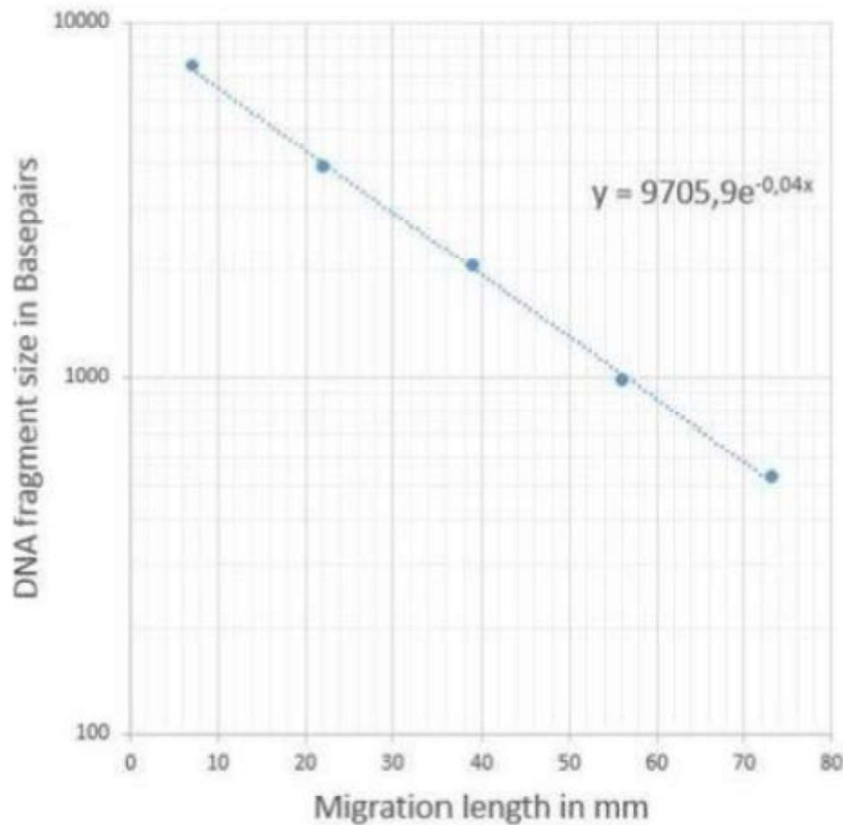
BglII one time - the two others two times

Explanation:

The number of times each restriction enzyme cuts the plasmid pRUT145 can be deduced from the number of bands produced after digestion. A plasmid is circular, so if an enzyme cuts it only once, it produces one linear DNA fragment. If it cuts twice, two fragments appear, and so on. In the provided gel, the BglII digest yields a single band, meaning it cuts only once. The ClaI and SnaBI digests each yield two bands, indicating that they cut the plasmid at two separate sites. This interpretation relies on the principle that the number of DNA fragments observed in a digest equals the number of cuts in a circular plasmid.

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B. Construct a standard curve, and calculate the size of all the plasmid fragments. (Hint: In Excel plot the known fragment sizes as function of the migration distance in a semi-logarithmic plot).



BglII: 7.000 bp; ClaI: 5750 + 1250; SnaBI: 4300 + 2700; SnaBI + BglII: 2700 + 2*2150;

ClaI + BglII: 5750 + 850 + 400

Explanation:

To determine the sizes of the unknown DNA fragments, a standard curve is created by plotting the logarithm of known marker fragment sizes against their migration distance in the gel. DNA separates by size during agarose electrophoresis, but the relationship is not linear. Instead, fragment migration distance is approximately inversely proportional to the logarithm of molecular weight. Because of this nonlinear relationship, direct measurement of distance cannot be used to estimate size—they must be interpreted using a standard curve. Once the curve is established, the sizes of unknown fragments can be accurately interpolated from their migration distances. This method relies on the predictable biophysical behavior of DNA during electrophoresis.

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C. What is the approximate size of pRUT145?

7.000 bp (7 kb)

Explanation:

The plasmid size is determined from the digest in which the enzyme cuts only once—in this case, BglII. A single cut linearizes the plasmid, and the resulting fragment represents the entire plasmid length. By comparing the migration distance of this fragment to the standard curve, the plasmid is calculated to be approximately 7,000 base pairs. This approach works because in a circular plasmid, one cut produces exactly one linear DNA molecule, whose size corresponds to the full plasmid.

D. Why does the *Sna*BI - *Bgl*II double digest only result in two bands?

The lower band is a double band 2 different DNA fragments with the same size. Note the intensity of the lower band is higher!!!

Explanation:

The *Sna*BI–*Bgl*II double digest should theoretically produce three fragments, because *Sna*BI cuts twice and *Bgl*II cuts once. However, the gel shows only two visible bands. This occurs because two of the fragments are the same size and therefore migrate together as a single band. These co-migrating fragments also produce a band with noticeably higher intensity because it contains twice the amount of DNA. This phenomenon exemplifies a limitation of agarose electrophoresis: fragments of identical or nearly identical size cannot be separated and appear as a single band.

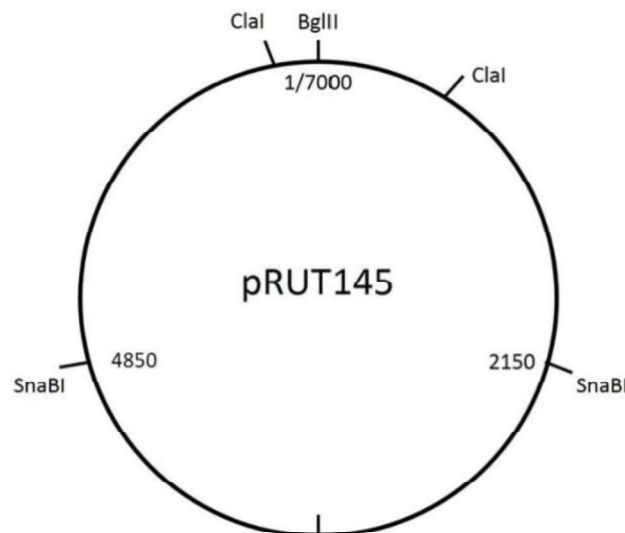
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E. Construct a restriction enzyme map of pRUT145. Set the BglII site to 1.

The restrict map of pRUT145 - Note that there are two different solutions for the ClaI sites

Explanation:

A restriction map is created by integrating the fragment sizes from single and double digests. The key idea is fragment additivity: fragments from double digests must match combinations of fragments from single digests and must sum to the total plasmid size. Because each enzyme has known cut frequencies and the total plasmid length is established, the relative positions of the restriction sites can be deduced by matching fragment sizes in a way that satisfies all observed patterns. This process reconstructs the circular arrangement of restriction sites, demonstrating how structural maps can be inferred from fragment data.



F. Suggest an alternative method to construct the map.

Sequence the plasmid and do a search for the restriction sites.

Explanation:

An alternative and more definitive method for constructing the restriction map is to sequence the plasmid. Once the full DNA sequence is known, restriction sites can be identified computationally by scanning for the nucleotide recognition sequences of the enzymes. This approach eliminates ambiguity, resolves fragments of identical size, and requires no gel-based interpretation. It reflects the broader principle that DNA sequencing has replaced classical mapping methods for nearly all modern applications due to its accuracy and completeness.

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Prep: læs læsestof, se videoer og andre dokumenter lagt op til ugen, lav ordbog for ugens ord

Forlæsning: noter?

Gruppearbejde: besvarelse af problems samt den korrekte besvarelse givet i solutions, tjek learning objectives og besvar?

Esperanza besvarer spørgsmål kl 8-9 fredage (learn)

Reeksamen: 1 uge i juni – mundtlige – 12 spg forud – open book

Molecular Biology Week 1: Course introduction and methods in molecular biology

Learning objectives

- Be able to use correct genetic nomenclature
- Describe the mode of action by restriction nucleases
- Propose a way to clone a DNA fragment into a plasmid vector
- Describe and analyse gel electrophoresis experiments
- Interpret simple Blotting experiments (Southern, Northern and Western)
- Describe DNA sequencing techniques: dideoxy- (or Sanger-) and next-generation sequencing. For the latter, a detailed understanding of the molecular principles is not

required.

- Design and explain a PCR experiment. This includes designing appropriate primers.
- Describe qPCR and Reverse Transcriptase PCR, and know the purpose of the different PCR techniques

Notes from the lecture

- 1 The central dogma

1. The central dogma



Difference between eukaryotes and prokaryotes central dogma: Slide 7-9

Transcription to translation: **RNA polymerase**

Ribosome: transform the mRNA into a protein

- 1.1 Prokaryotes
- 1.2 Eukaryotes
- 2 Molecular biology usages

Usages: Slide 10

- 3 Cloning a protein for heterologous production

heterologous production: slide 11

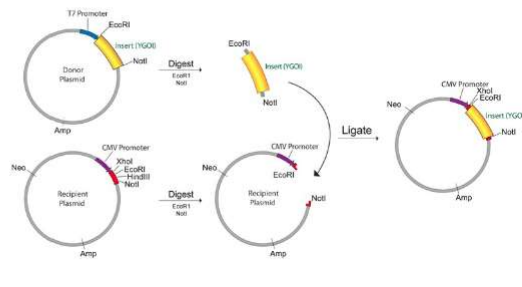
vectors: a way to get something to a different place, a way to transfer DNA to specific organisms.

- 3 Cloning a protein for recombinant production

Recombinant production: slide 12

3 Cloning a protein for recombinant production

- Choose the gene (insert) and the vector
- Amplify the insert by PCR
- Check the size by electrophoresis
- Digest vector and insert with compatible restriction enzymes
- Purify the fragments
- Ligate vector and insert
- Transform competent cells and select transformants
- Pick a single colony, extract the plasmid DNA and send for sequencing



– 3.1 Obtaining the gene

For prokaryotes:

Plasmid: a plasmid is a small, circular piece of DNA that separates from the main DNA in a cell, they carry:

- Origin of replication (every time the genome of the bacteria will copy, the vector will also copy)
- Selection marker (antibiotic resistance for an example, whatever does not have the wanted gene dies)
- Multiple cloning site
- If for expression, promoter and terminator (for binding etc)

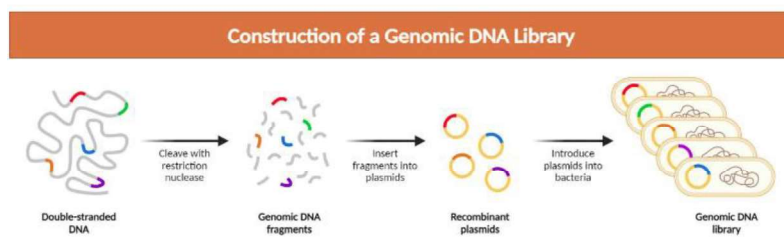
For bacterial plasmids: (difference??) slide 13

Database

- Genomic DNA

If you don't know the sequence: break RNA randomly, clone and make a library:

A library: A library is a group of many different DNA or RNA molecules, stored in a way that allows researchers to search through them to find specific genes, sequences, or functions.



“make a library and screen for what?”

- mRNA

obtaining mRNA: difference between prokaryotes and eukaryotes:

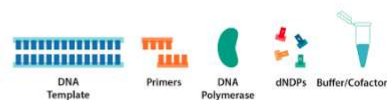
3.1 Obtaining a gene: mRNA

Do you know the sequence?

- Yes - design primers and amplify using RT-PCR
- No –
- In eukaryotes use a polyT to catch the polyA tail and a random primer on the other end and make a library
- In prokaryotes use random primers and make a library
 - Reverse transcriptases (also called **retrotranscriptases**) are enzymes made by some viruses that can copy RNA into DNA.
 - 3.2 PCR

3.2 PCR

- PCR: polymerase chain reaction. Nobel prize 1993
- It consists of using a heat-resistant polymerase to amplify DNA
- Ingredients:
 - Template DNA
 - Flanking oligonucleotides matching the sequence
 - dNTPs
 - Polymerase
 - Buffer
 - Heat resistant DNA polymerase
 - Thermocycler



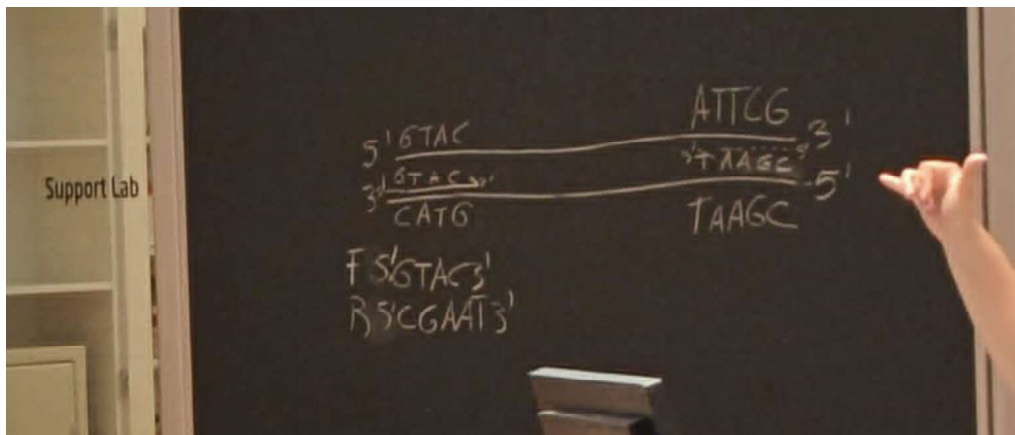
PCR calculations:

Imagine you have 20 copies of template in your mix.
How many copies do you have after 20 PCR cycles?

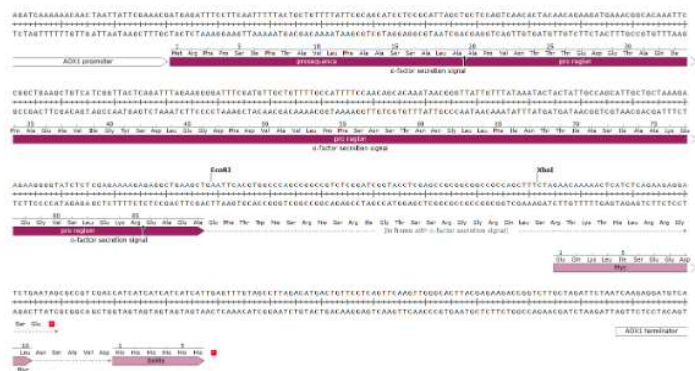
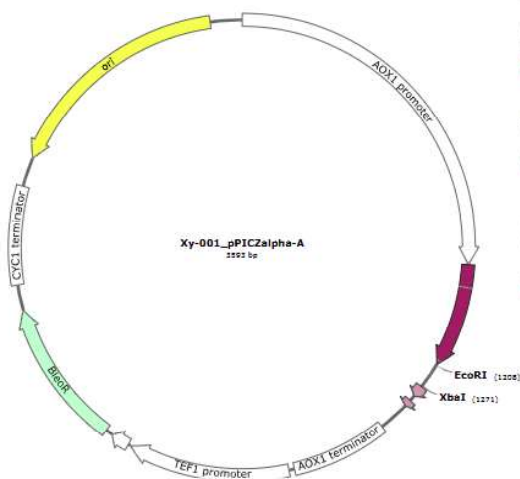
Final copies = $N_0 \times 2^n$
 N_0 = starting number of DNA copies
 N = number of PCR cycles

Primers: Primers are part of the final sequence which allows for adding sequences on the DNA edges (for example restriction sites) – one of the primers are always gonna look exactly like the opposite sequence – the 5' end meeting the 3' end that the primer needs to attach itself to. You will need two primers, one for each end strand in the doublestranded DNA.

Always set in 5' to 3' direction. (the writing direction of the sequence)



3.2 PCR



EcoRI – GAATTC
XbaI – TCTAGA

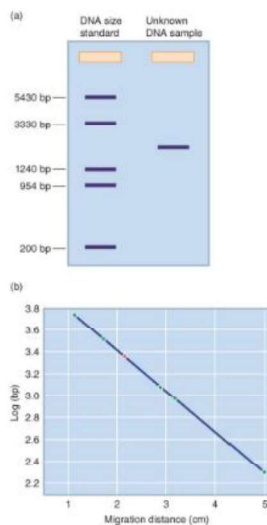
Design two primers of 25bp to amplify Pru p 3 sequence adding EcoRI in the 5' and XbaI to the 3'

- 3.3 Electrophoresis- a separation technique focusing on size

DNA always have a negative site.

Electrophoresis is a laboratory technique used to separate charged molecules—such as DNA, RNA, or proteins—based on their size, charge, and shape by applying an electric field through a gel or other medium.

- Bigger molecules migrate slower (stay closer to the wells). The migrations distance is related to the log(bp)
- Shape also matters. A dsDNA circular DNA is supercoiled and has an apparent migration that is faster than is actual shape.
- Ladders are usually linear pieces of DNA so they can only be compared with linear DNA



Migration always depends on the size, the charge and the shape! – circular DNA that is very t  t-pakket will run faster through gel electrophoresis than the exact same protein that is stretched out would – see slide25.

Promoter: piece of DNA the place the the rna-polymerase is most likely to bind to, so the function of a promoter in an **expressive vector** is to initiate transcription

– 3.4 Northern and Southern blotting: slide 35

You know the sequence but it is a very complex mix: imagine that you are working with cancer cells and you want to stop a specific gene from working. (inhibition)

You have to double check that the gene are blocked or not.

If you know the sequence you can use blotting. Used to find one specific sequence among many others.

3.4 Northern and Southern Blotting

- Detection of specific known sequences
- Steps:
 - Run an agarose gel to separate fragments regarding size
 - Transfer to a membrane.
 - Incubate the membrane with a hybridization probe: a piece of DNA which sequence is complementary to the sequence we want to detect and is fluorescent or radiatively labelled
 - Visualize
 - The name depend on the nature of the molecular we are analyzing



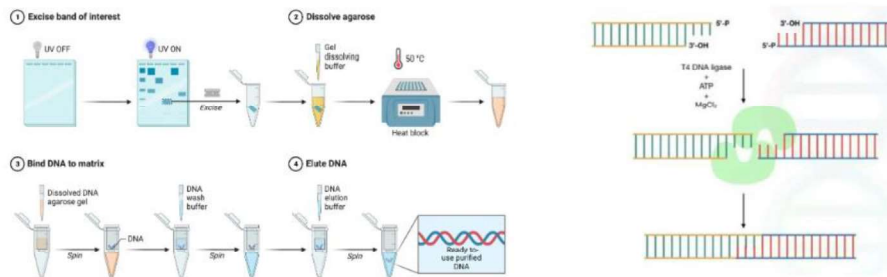
Involves – electrophoresis, tranfers it and then something binds to it and vislualizes it. (?)^^

Slide 36^^

– 3.5 Restriction and ligation

3.5 Restriction and ligation

- Cut both insert and vector with compatible restriction enzymes
- Purify the fragments using gel purification
- Perform ligation with T4 ligase



Specific sequences that a palindromic like GAATTC (CTTAAG).

Restriction enzyme: restriction endonuclease

– 3.6 Transformation and selection, Slide 39

Competent cell: A competent cell is a bacterial cell that is able to take up foreign DNA from its environment.

Selection markers:

Selection. It requires that the “wild type” cells cannot grow in certain conditions unless it is transformed

- Antibiotic resistance
- Plasmid contains a gene that confers resistance to antibiotic
 - Auxotrophy supplementation
- Plasmid contains an enzyme involved in an essential amino acid synthesis
 - Blue-White colony screening (slide42)
- Differentiates between positive transformation and correct insertion
 - 3.7 Sequencing

Sanger sequencing : def ?

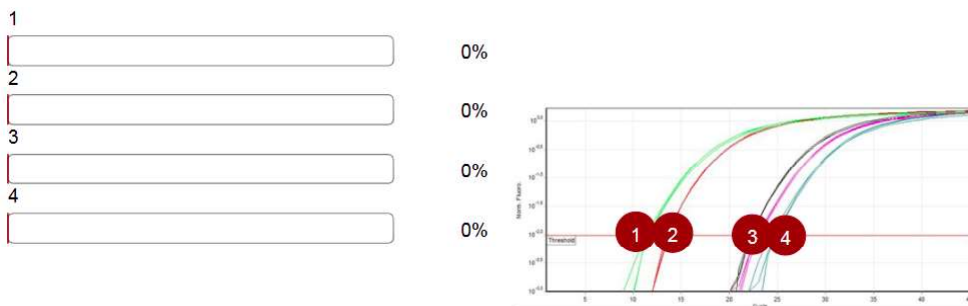
- 3.8 DNA quantification
- UV/Vis / Fluorecence

- qPCR, quantitative PCR (opposite direction of the slide 21) – you can see the amount of fluorescence after x amount of cycles and then you can go back and see how much DNA you had in the beginning.

Why is codon optimization used in heterologous protein expression? – To match codon usage to the host organism.

What does the A260/A280 ratio help determine in DNA quantification? – when you have a higher A280 than A260 means that you have more protein than you should - Protein or RNA contamination

Which DNA is on a higher concentration?



The answer is 1. **Why?**

find ud af. Ecor1 og xba1, vektorer, blotting, sequencing (terminator), GC content

Molecular Biology Week 2: Transcription in pro- and eukaryotes

Learning objectives

Learning objectives:

After today's lecture and exercises, a student who has met the objectives will be able to:

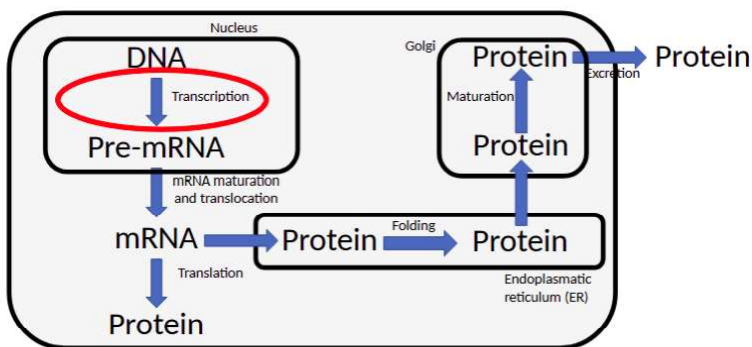
- Describe the biochemical reaction in transcription – enzymes, substrates and products
- Distinguish between the template and coding DNA strands and describe RNA strand polarity (5' to 3').
- Describe the initiation, elongation, and termination processes
- Explain what a promoter is
- Know the structure and subunits of the prokaryotic RNA polymerase – the core enzyme and the holoenzyme
- Identify the consensus sequence of prokaryotic and eukaryotic household promoters

- Describe the interactions between the prokaryotic Sigma factor and specific DNA sites (-35 and -10)
- Explain in your own words how intrinsic termination works in prokaryotes
- Know that RNA is transcribed by three different RNA polymerases in eukaryotes and what kind of RNA they produce
- Describe the role of the TATA box and TBP in eukaryotic transcription initiation
- Describe the sequential process in eukaryotic transcription initiation of all three RNA polymerases
- Evaluate how promoter engineering can improve gene expression in biotechnology.

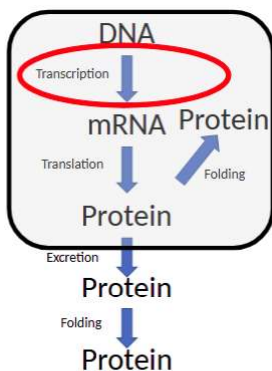
Notes from the lecture

1. Introduction

Central dogma in eukaryotes:

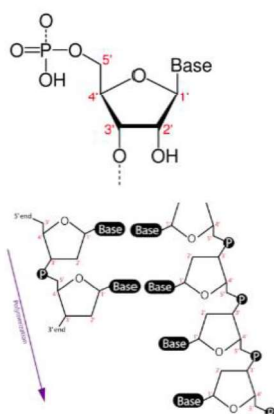
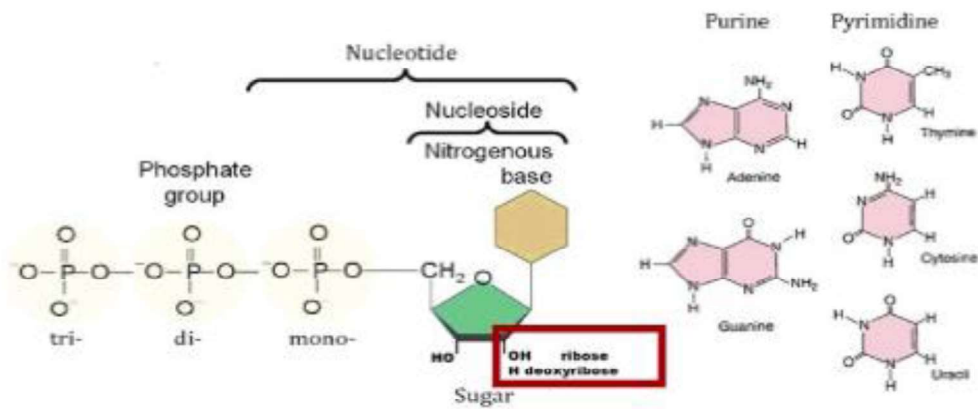


Central dogma in prokaryotes:



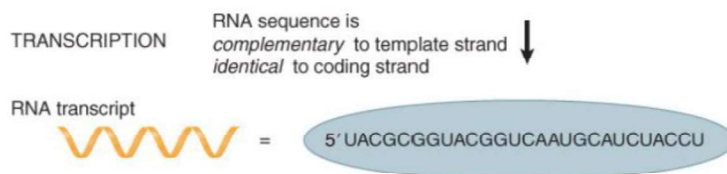
- DNA is made of deoxyribonucleotides
 - Building blocks are Deoxy-Ribonucleoside-Tri-Phosphate
- dATP, dGTP, dCTP, dTTP
- RNA is made of ribonucleotides
 - Building blocks are Ribonucleoside-tri-phosphate

- ATP, GTP, CTP, UTP
 - Nucleotides are always polymerized in 5' to 3' using the 3' to 5' strand as template

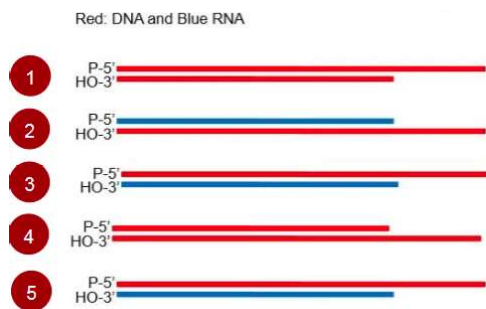


3' and 5' gives the direction of the sugar base

- All known DNA and RNA polymerases add nucleotides to the 3' end.
- Nucleotides are always polymerized in 5' to 3' using the 3' to 5' strand as template
- Coding strand (non-template strand) – The DNA strand that has the same sequence as the mRNA



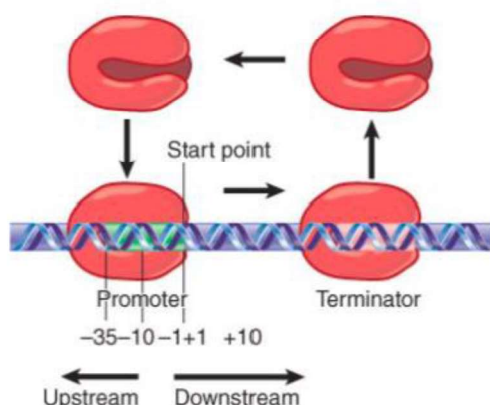
Which of the nucleic acids can be elongated by a DNA polymerase?



Answer: 4

2. Transcription in prokaryotes

- 2.1 The RNA polymerase
- Prokaryotic transcription is the process of creating a messenger RNA (mRNA) copy of a gene in prokaryotic cells. This process occurs in the cytoplasm. Key features include the involvement of RNA polymerase and the absence of a nuclear membrane, allowing for simultaneous transcription and translation.
- The transcription process has three steps:
 - Initiation
 - Elongation
 - Termination
- **Promoter** – A region of DNA where RNA polymerase binds to **initiate transcription** at the start site – The position on DNA(+1) corresponding to the first base incorporated into RNA.
- The RNA is synthesized in the **elongation** stage
- **Terminator** – A sequence of DNA that causes RNA polymerase to terminate transcription



Upstream: sequence is to the left

Downstream: sequence is to the right

- Core enzyme – The polymerase multisubunit complex of about 400 kDa with the general structure $\alpha_2\beta\beta'\omega$
- However, the RNA polymerase alone is not enough to initiate transcription
- The RNA polymerase form that is competent to initiate transcription is known as holoenzyme: the five subunits of the core enzyme and (sigma) σ factor. (the protein that tells where the polymerase has to bind)

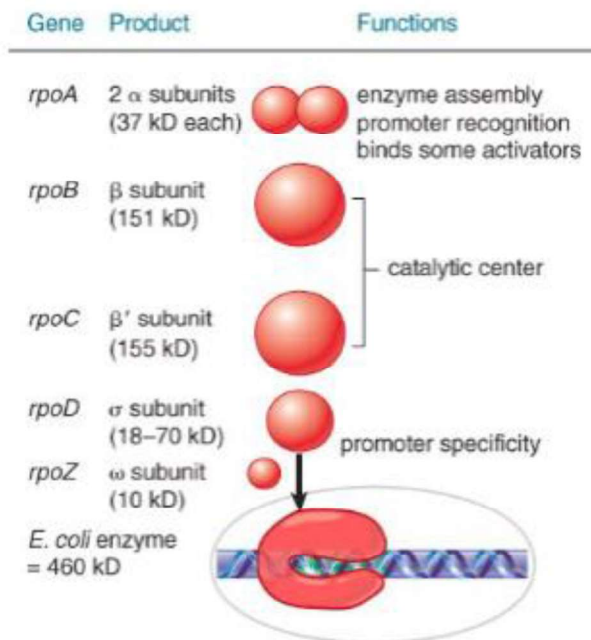
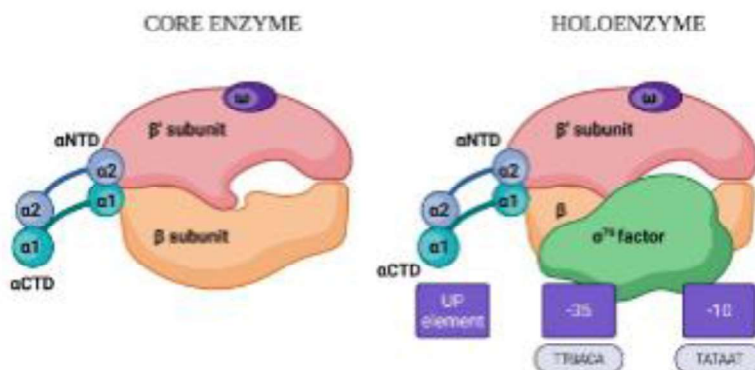
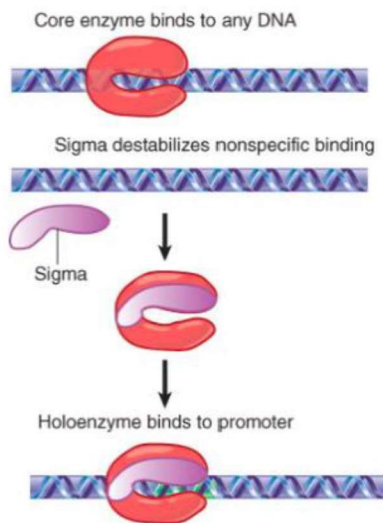


FIGURE 15-10

Core enzyme and holoenzyme:



- Sigma factor changes the DNA-binding properties of RNA polymerase so that its affinity for general DNA is reduced and its affinity for promoters is increased.
- There are different sigma factors that can modulate which “set” of genes are expressed depending on environmental signals (i.e., under a heat-shock) – more in the prokaryotic gene regulation.



– 2.2 Promoter structure

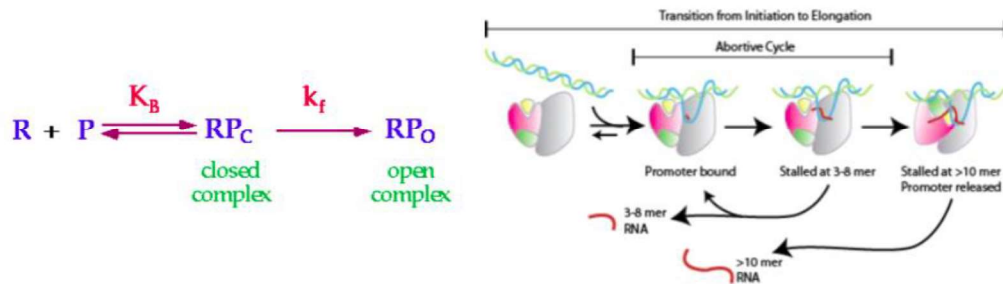
- The promoter consensus sequences usually consist of a purine at the start point (+1), a hexamer with a sequence close to TATAAT centered at about –10 (–10 sequence or Pribnow box), and another hexamer with a sequence similar to TTGACA centered at about –35 (–35 sequence).
- Individual promoters usually differ from the consensus at one or more positions.
- The distance between –35 and –10 is always 17 +/- 1 nucleotides
- The consensus sequence is not absolutely conserved
- The –10 sequence is NOT a TATA box. TATA box is for eukaryote

```

5' -CTGTTGACAATTAATCATCGAACTAGTTAACTAGTACGCAAG-3'
5' -CTATTGCTGTACGCGGAGCATGCATTAGAGTTACGATCAGC-3'
5' -TGGTTCCAAATCTACTACGTACGTATATACTCGATCTCGGA-3'
5' -TTTGTGATCTGAGCATGCATGCATGCTCTAATCACTGTCTAG-3'
5' -TAGTTGCATGGAGAGCATCGATCGATTAAAATGCTGACTAGG-3'
5' -TTCCTGACACGGCTAGATGCATGCATTATAATTCTATGCCCG-3'
    -35 sequence          Pribnow box          +1
                        (-10 sequence)      Transcr. start
Consensus: TTGACA-----N17-----TATAAT-N7-----A

```

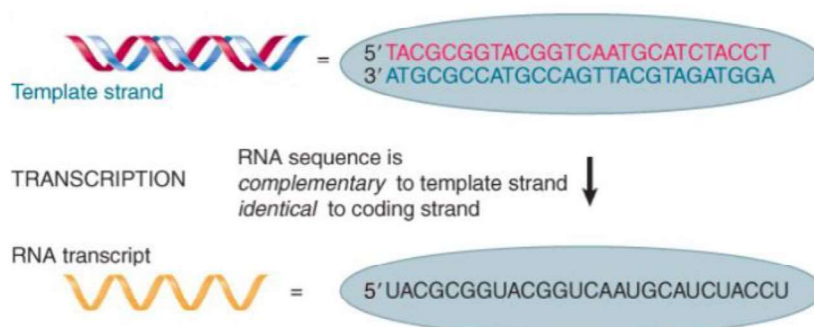
- The nucleotides outside the consensus (which kind and how many) are important for promoter strength



2.3 Initiation: written summary

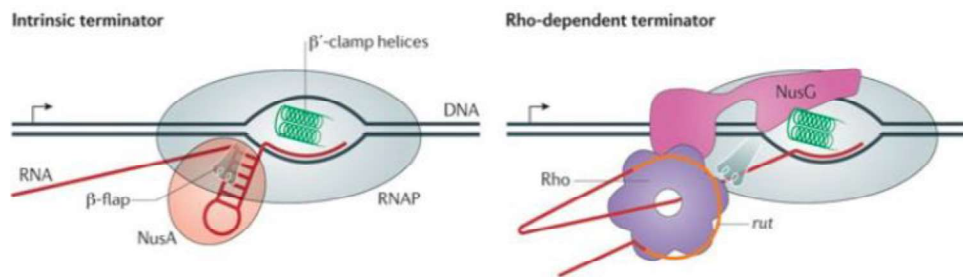
- The RNA polymerase holo-enzyme slides along the DNA molecule
- The RNA polymerase binds to the promoter primarily by interaction between region 4.2 in the sigma factor and the – 35 sequence
- Additional contacts between the enzyme and the DNA stabilizes the complex including interaction between the region 2.4 in the sigma factor and the – 10 sequence
- The DNA melts facilitated by the affinity between the region 2.3 in the sigma factor and the – 10 sequence in the non-template (coding) strand
- The first phospho-diester bond is formed in the active site
- The interactions between the sigma factor and the DNA must weaken in order for the elongation to proceed
- A cycle of abortive initiations may occur before successful promoter clearance
- Upon successful promoter clearance the sigma factor released, and the core enzyme synthesizes full length RNA
 - 2.4 Elongation

- Transcription occurs in a 5' to 3' direction using the 3' to 5' strand as template.
- Nucleotides are added to the 3'OH end of the growing RNA chain



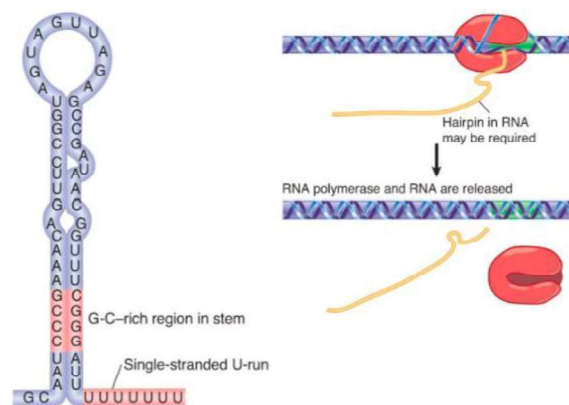
– 2.5 Termination

- Two classes of terminators have been identified:
 - **Intrinsic terminators:** Those recognized solely by RNA polymerase itself without the requirement for any cellular factors are usually referred to as
 - **Factor (or rho)-dependent terminators:** require cellular proteins eg. Rho



Intrinsic terminator:

- Intrinsic terminator:
 - The nascent mRNA forms a hairpin followed by a polyU sequence. The hairpin is identified within the RNA polymerase and the RNA polymerase detaches
 - The signals for termination lie within sequences *already transcribed* by RNA polymerase



- 2.6 Engineering potential

2.6 Engineering potential

- Let's take the Phytase example back. You want to investigate why the *appA* gene expression is low in wild type.
 - Enter in the published genome of *E. coli* K12 in NCBI Genomes https://www.ncbi.nlm.nih.gov/nucore/NC_000913.3?report=graph
 - Use the "Find" bar to look for "appA" and access it
 - Click on the "ATG" symbol in the navigation bar to zoom for sequence
 - Try to find out the promoter upstream the coding sequence and discuss your findings in pairs



The issue is that the promoter is not perfect – it is found a bit far from the -35 sequence:

Which parameters would you vary on your library and how would you test the promoter strength?: Make a library of DNA that would have different sequences that looks more like the

consensus in front of the (?) – change promotor and see if expression level changes – test the minus 10 to 25.

Which of the following is essential for the initiation of transcription in bacteria?:

The sigma spliceosome		0%
Poly-A tail addition		0%
Sigma factor binding to the promoter		0%
RNA polymerase binding to the 5' cap of mRNA		0%

Answer: The sigma factor and sigma factor binding to the promoter

A promoter is a __ and an RNA polymerase is a __, whereas ribosomes are made up of __:

DNA sequence, Protein, tRNA		0%
Protein, Protein, DNA sequences		0%
RNA sequence, DNA sequence, mix of RNA and proteins		0%
DNA sequence, protein, mix of RNA and protein		0%

Answer is the last one – DNA sequence, protein and mix of RNA and protein.

In bacteria, which of the following is a key feature of intrinsic (Rho-independent) transcription termination?:

Formation of a hairpin loop followed by a poly-U sequence in the RNA		0%
Binding of the ribosome to the mRNA		0%
Degradation of the RNA by RNase		0%
Addition of a 5' cap to the RNA		0%

Answer is the first one – formation of a hairpin loop followed by a poly-U sequence in the RNA.

3. Transcription in eukaryotes

– 3.1 The RNA polymerases

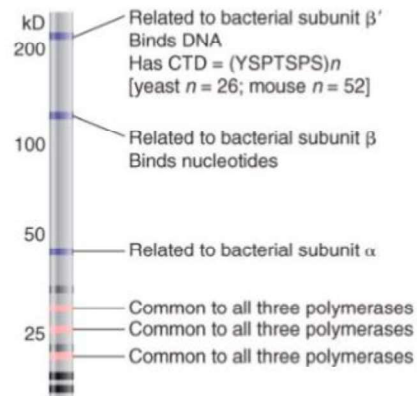
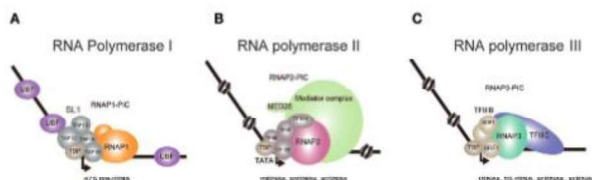
anti-apoptotic genes: ?

consensus sequences: ?

RNA polymerase II is used by eukaryotic cells to produce mRNA. We will therefore focus in this today.

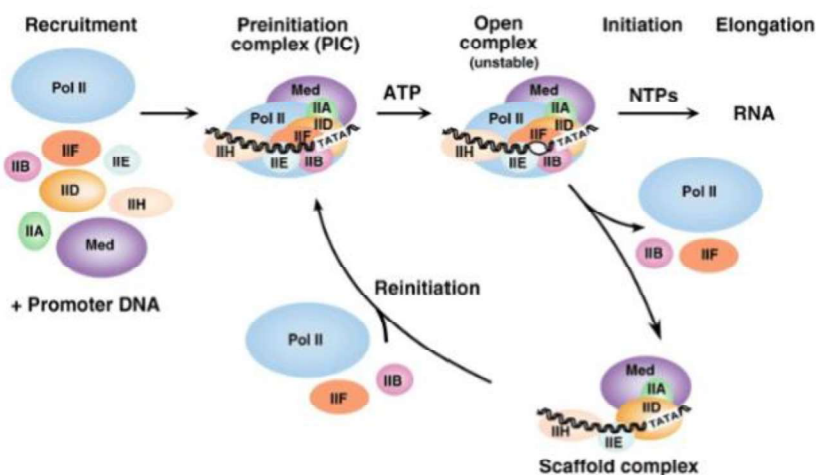
3.1 RNA Polymerases

- **RNA polymerase I** synthesizes rRNA in the nucleolus
- **RNA polymerase II** synthesizes mRNA in the nucleoplasm
- **RNA polymerase III** synthesizes small RNAs (often regulatory) in the nucleoplasm
- All eukaryotic RNA polymerases have about 12 subunits.
- Some subunits are common to all three RNA polymerases.
- The largest subunit in RNA polymerase II has a **carboxy-terminal domain (CTD)** consisting of multiple conserved repeats of a heptamer.



3.1 RNA Polymerases

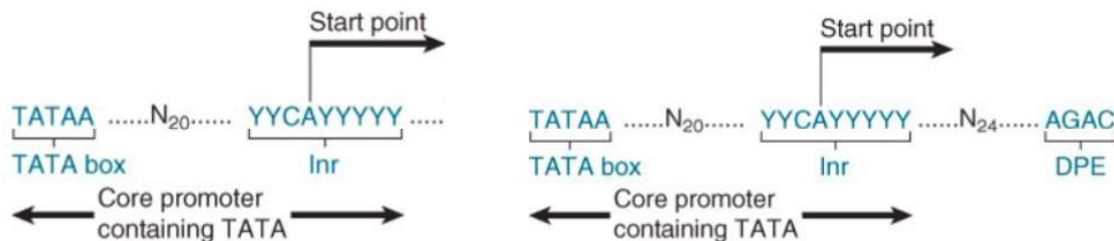
- We will focus on RNA polymerase II because it is the enzyme responsible for the transcription of all mRNA in eukaryotes



– 3.2 Promoter structure

3.2 Promoter structure

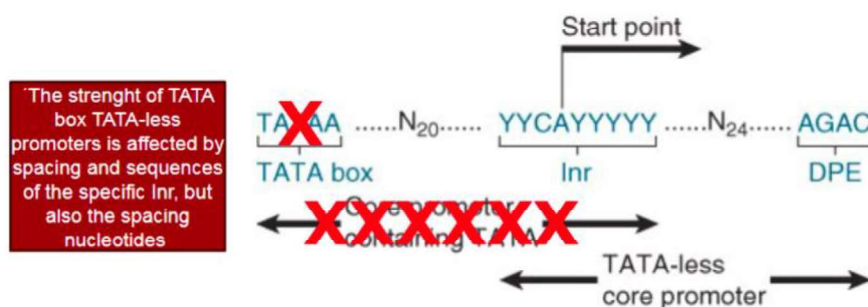
- RNA polymerase II promoters frequently have a short, conserved sequence Py₂CAPy₅ (the **initiator Inr**) at the start point.
- The **TATA box** is a common component of RNA polymerase II promoters: it consists of an A-T-rich octamer located approximately 25 bp upstream of the start point.
- They may contain or not a **downstream promoter element (DPE)**



TATA-BOX

3.2 Promoter structure

- Some RNA polymerase II promoters do not contain a TATA box (**TATA-less promoters**).
- The **downstream promoter element (DPE)** is a common component of such promoters
- A core promoter for RNA polymerase II includes the **Inr** and, commonly, either a **TATA box** or a **DPE**.



– 3.3 Initiation

First part: Chromatin packing (slide 42)

Second part: TBP (TATA binding protein) – the RNA polymerase is “blind” w/o the TATA binding protein will bind to the promoter, and then it gets associated with different proteins and they form a big mole (?). even for TATA-less promoters TBP is still needed (slide 43).

TBP binds to TFIID and it bends it (?) – slide 46

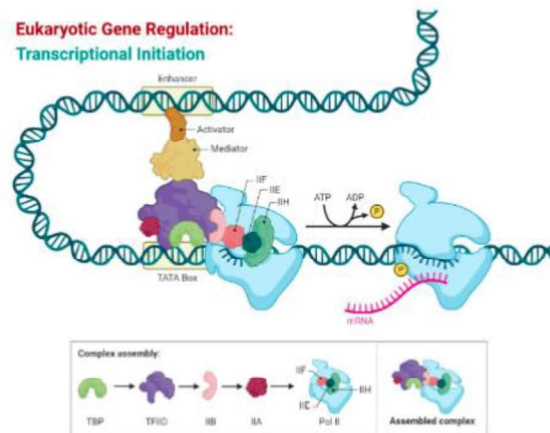
Basal apparatus: slide 48

- 3.4 Elongation
- 3.5 Termination
- 3.6 Engineering potential

3.6 Engineering potential

- You're designing a strategy to reduce BCL2 expression in cancer cells by interfering with its **transcriptional initiation**—without editing the coding sequence of BCL2
- BCL2 is a gene that helps cells resist apoptosis (programmed cell death). It's often overexpressed in cancer, allowing abnormal cells to survive chemotherapy. This makes BCL2 a target for therapies aiming to reduce its expression.
- Transcriptional initiation of BCL2 is regulated by:
 - A GC-rich promoter (no TATA box)
 - Binding of transcription factors like SP1, E2F, and NF- κ B (protein that interact with the basal apparatus and increase the recruitment of the RNA pol II to the promoter)
 - Chromatin state and histone modifications

Propose two strategies for reducing BCL2 expression affecting the transcription of the gene.



- Answer: CpG islands (CGIs) are GC-rich regions of DNA, typically found near gene promoters, with a high frequency of CpG dinucleotides. They play a crucial role in regulating gene expression; when CGIs are unmethylated, they promote transcription, but dense CpG methylation silences the associated gene by altering chromatin structure to restrict access for transcription machinery. – so the answer is that you can methylize the GC-rich promoter, altering the chromatin structure hindering the first step of translation in eukaryotes – chromatin packing.
- Answer: alter to add terminator before gene location of BCL2
- Answer: if you make the sequence to get tighter around the histones then the ? cant bind at all

Which element is shared by all eukaryotic RNA polymerases during initiation of transcription:

The TATA-Box

0%

The TATA-box binding protein (TBP)

0%

The sigma factor

0%

The initiator element (Inr)

0%

Answer: The TATA-binding Protein (TBP)

Which trait is shared by eukaryotic and prokaryotic RNA polymerases:

The holo-enzyme can alone initiate transcription		0%
The catalyse the formation of phosphodiester bonds in RNA		0%
The are producing primers in replication		0%
They are both dependent on transcription factors (TF)		0%

Transcription factors are only used in eukaryotes

Answer: They catalyze the formation of phosphodiester bonds in RNA.

What is the primary function of the transcription factor TFIID in eukaryotic transcription initiation

It unwinds the DNA double helix to allow RNA polymerase II to bind		0%
It phosphorylates RNA polymerase II, initiating elongation		0%
It binds to the TATA box, helping recruit RNA polymerase II to the promoter		0%
It catalyses the removal of introns from the pre-mRNA		0%

Answer: it binds to the TATA box, helping recruit RNA polymerase II to the promoter.

4. Comparative overview

Feature	Prokaryotes	Eukaryotes
Location	Cytoplasm	Nucleus
RNA Polymerase(s)	One RNA polymerase	Three RNA polymerases: • Pol I → rRNA • Pol II → mRNA, some snRNA • Pol III → tRNA, 5S rRNA
Promoter Elements	-35 and -10 boxes (e.g., TTGACA, TATAAT)	TATA box, Inr, DPE (varies by gene)
Initiation Factors	Sigma factor (σ) required for promoter recognition	General transcription factors (e.g., TFIID, TFIIE, TFIIIB, TFIIH) assemble into Pre-Initiation Complex
Initiation Site Recognition	Sigma factor directly recognizes promoter	TFIID (with TBP) binds TATA box; other GTFs recruit RNA Pol II
Elongation	Core enzyme synthesizes RNA after σ dissociates	Requires elongation factors; Pol II CTD gets phosphorylated to proceed
Termination	Rho-dependent or intrinsic (hairpin + U-tract)	Linked to RNA cleavage and polyadenylation; involves Xrn2 (torpedo model) and/or Pol II conformational change (allosteric model)
Coupled with Translation?	Yes – translation begins while mRNA is still being transcribed	No – transcription and translation are spatially separated
RNA Processing	Minimal; rRNAs and tRNAs processed, mRNAs usually not	Extensive: 5' cap, splicing, 3' poly(A) tail added before export
Chromatin Structure	Absent	Transcription must overcome chromatin/histone packaging; epigenetic regulation
Engineering Handles	Promoter/operator design, sigma factor tuning, terminator modification	Synthetic promoters/enhancers, chromatin modifiers, transcriptional repressors (e.g., dCas9-KRAB)

3.4) Termination

- RNA polymerase I terminates at specific sites and is dependent on a specific factor
 - (TTF1)
- RNA polymerase II termination is closely linked to mRNA processing (next lecture)
- RNA polymerase III appears to terminate at specific sites with a oligo-dT sequence on the
 - coding strand (similar to procaryotes)

Molecular Biology Week 3: RNA Maturation and decay

Learning objectives

In this lecture, you will explore the critical steps of RNA post-transcriptional processing, a process that dramatically increases the complexity and regulation of gene expression in eukaryotic cells. Once transcription produces a pre-mRNA, it is not immediately ready for translation. Instead, it must undergo several tightly controlled modifications:

- Addition of a 5' cap
- Splicing to remove introns and join exons
- Addition of a poly(A) tail at the 3' end
- Transport of the mature mRNA from the nucleus to the cytoplasm

These modifications are not just technical necessities—they are regulatory checkpoints that influence whether, when, and how much protein is produced. Cells also use these processes to detect and eliminate defective RNAs, ensuring accuracy and efficiency.

Understanding this material is essential for both fundamental biology and applied biotechnology. For instance, mRNA stability and splicing are actively engineered in mRNA-based vaccines and in the development of therapeutic proteins. You'll see examples of how mRNA half-life can be tuned to affect gene expression outcomes—knowledge that is already being used in synthetic biology, and drug design.

After today's lecture and exercises, a student who has met the objectives will be able to:

- Explain how to process pre-mRNA to mature mRNA
- Identify what is an intron and an exon, and how are they identified in pre-mRNAs
- Explain, mechanism and role of 5'-capping
- Identify splice sites for excision of introns
- Describe the steps are involved in splicing, including formation of a lariat structure
- Describe the spliceosome and function of the snRNPs U1, U2, U5, U4 U6 and other factors involved in RNA splicing
- Explain in your words the stages of splicing, E, A, B1, B2, C1 and C2 complexes
- Understand and describe the importance and mechanisms of alternative splicing

- Explain the mechanism and role of poly-adenylation
- Determine the half life of an mRNA based on experimental data
- Describe mRNA degradation mechanisms in prokaryotes and eukaryotes
- Propose means to stabilize mRNA and avoid degradation (prokaryotic and eukaryotic)

Notes from the lecture

1. Introduction - The fate of bacterial and eukaryotic RNA after transcription

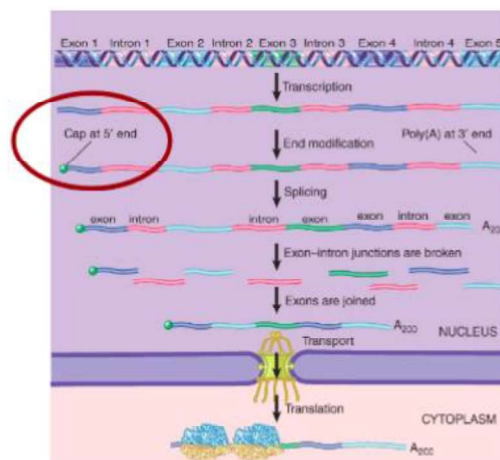
In prokaryotes everything happens at the same time, but in eukaryotes the transcription happens in the nucleus and then the mRNA has to be transported out of the nucleus to the ribosomes in the cytoplasm for the process to move on.

In order for the mRNA to come out to the cytoplasm certain structures need to be added to the 5' and 3' end of the mRNA, because the nucleus is not porous and therefore it can't just diffuse out into the cytoplasm by itself.

2. 5'-capping

2. 5'-capping

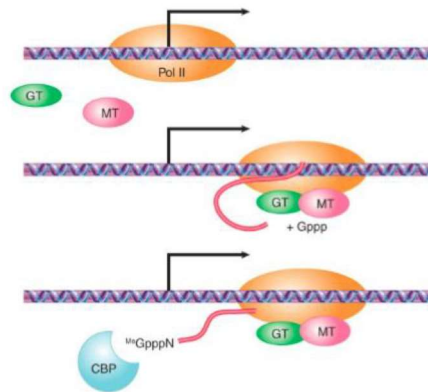
- 5'-capping is a modification of the 5' end of mRNA adding a G via 5'-5' link catalysed by the enzyme guanylyl-transferase.
- It is not an ordinary 3'-5' phosphodiester bond and therefore the regular enzymes that degrade mRNA cannot attack.
- It happens when the RNA polymerase has transcribed around 30 nucleotides. It avoids that the mRNA is degraded from the free 5' edge.



As long as the mRNA is long enough (at least 30 nucleotides long) 5' capping can happen.

2. 5'-capping

- The capping process may be important for release from pausing of transcription.
- The Guanylyl-transferase (GT) and 7-Methyl-transferase are associated with RNA Pol II
- It happens when the RNA polymerase has transcribed around 30 nucleotides
- CBP: CAP binding protein. This protein binds to the 5'-cap and further protects the mRNA from degradation
- The cap structure is recognized by additional protein factors to influence mRNA stability, splicing, export, and most important translation



3. Splicing: introns and exons

3. Splicing

- A 5' cap is formed by adding a G to the terminal base of the transcript via a 5'–5' link catalyzed by the guanylyl-transferase
- The 5' cap of mRNA is methylated by 7-methyl-transferase (MT)
- Additional 5' cap methylation occurs in the cyto-plasm of small, non-coding RNAs
- The 5' cap of most mRNA is monomethylated, but some small noncoding RNAs are trimethylated.
- The additional methylations prevents the RNA from being translated

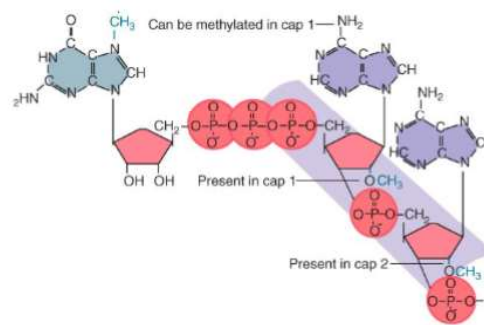


Figure 19.2: The cap blocks the 5' end of mRNA.

The cap is added at the three phosphates. On top of the caps there can be methylations and different methylations can be important for how it is translated.

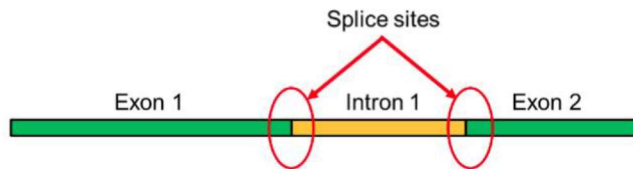
Splicing: the process in which the exons are excised from the pre-mRNA

Most of the mRNAs in eukaryotes ... splicing?

3.1 Splice sites

3.1 Splicing: splice sites

- Splice sites are the sequences immediately surrounding the exon–intron boundaries.
- At least four distinct classes of introns have been identified



- The four major intron classes:

3.1 Splicing: splice sites – the four major intron classes

- Group I introns: Self-splicing ribozymes that are found in rRNA, tRNA and mRNA
- Group II Introns: Also self-splicing ribozymes – found primarily in bacteria and organelles in rRNA, tRNA and mRNA
- Nuclear pre-mRNA Introns: They are found in the nucleus in protein-coding genes that are removed by spliceosomes
- Transfer RNA introns: They are found in tRNA genes and need proteins (enzymes) to be removed

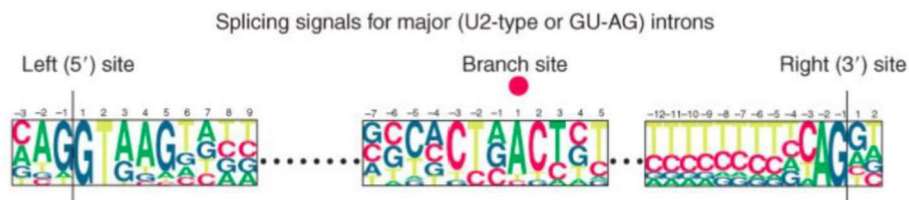
3.1 Splicing: splice sites

- The U2-type intron is the most abundant
- The 5' splice site at the 5' ("left") end of the major intron (U2-type) includes the consensus sequence GU.
- The 3' splice site at the 3' ("right") end of the major intron (U2-type) includes the consensus sequence AG.
- U2-type introns are also known as GU-AG introns (or in the past as GT-AG introns)



3.1 Splicing: splice sites

- The **GU-AG rule** (originally called the *GT-AG rule*) describes the requirement for these conserved dinucleotides at the first two and last two positions of introns in pre-mRNAs.
- The branch site is where the 5' end of the intron forms a phosphodiester bond, creating the lariat structure
- Notice that the recognition sites are inside introns (will be removed)

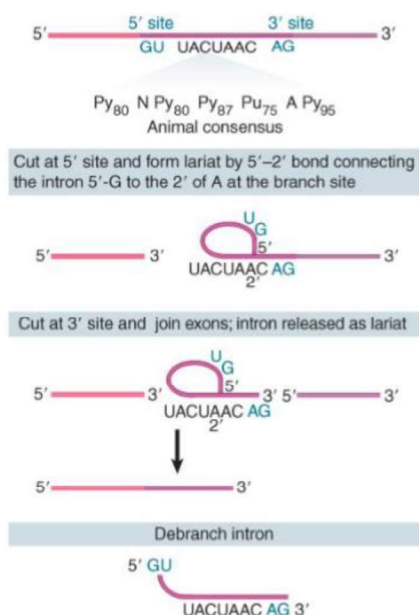


You can identify an intron with this format where if the letters GT and AG occurs 100% of the time (sequence conservation) and therefore it is very clearly the intron consensus sequences.

- The pre-mRNA splicing proceeds through a lariat (lasso)
- Splicing required the 5' and 3' splice sites and a branch site just upstream of the 3' splice site.
- A lariat is formed when the intron is cleaved at the 5' splice site, and the 5' end is joined to a 2' position at an A at the branch site in the intron

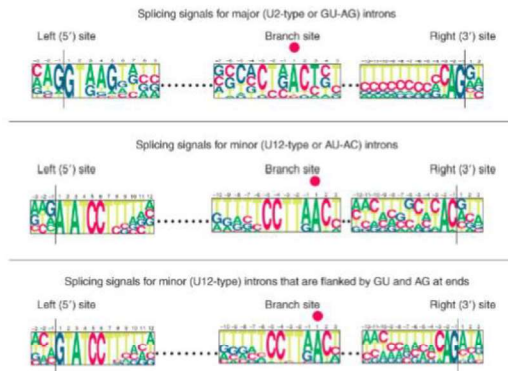
3.1 Splicing: splice sites

- An intermediate in intron removal is the formation of a lariat when it is cleaved at the 5' splice site and then ligated to the branch site in a 2'-5'-di-esterbond.
- The intron is released as a lariat when it is cleaved at the 3' splice site, and the left and right exons are then ligated together.
- Finally, the lariat is resolved and the intron RNA is then prone for degradation



3.1 Splicing: splice sites

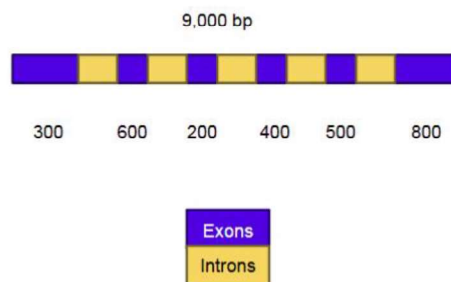
- Minor introns exist relative to the major U2 introns with the **GU-AG** sites
- Minor introns (e.g. U12 type) follow a general **AU-AC** rule with a different set of consensus sequences at the exon–intron boundaries.
- They always branch on an A



What is it that cannot be free?

Let's do it together

- A eukaryotic pre-mRNA is 9,000 bp long. It includes:
 - 6 exons: 300 bp, 600 bp, 200 bp, 400 bp, 500 bp, 800 bp
 - 5 introns between them
- Calculate the size of the mature mRNA and estimate the number of amino acids of the translated protein.
- If a poly(A) tail of 250 bp is added, what is the final size?
- If splicing removed exon 4 in an alternative isoform, how would the mRNA size and protein change?



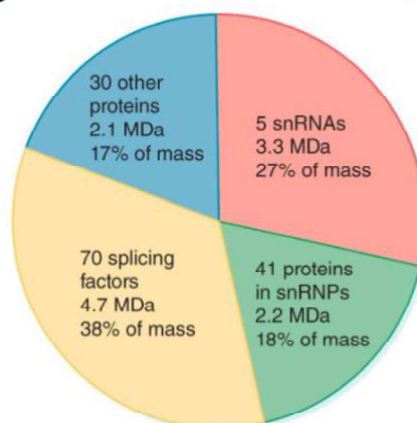
Mature mRNA: 2800 bp = 930 aa (aminoacids) + poly(A) tale is 3050 bp

3.2 The spliceosome

The splicing in an eukaryotic cell

3.2 Splicing: the spliceosome

- The spliceosome is a complex composed by proteins, small nuclear RNA (snRNAs), and small nuclear ribonucleoproteins (snRNPs) that controls and catalyses the mRNA splicing events.
- The RNAs are forming the catalytic site in the transesterification process (the formation of a phosphodiester bond between the 5' of the splice site and the 2' of the A in the branch site).
- The five snRNPs involved in splicing are U1, U2, U5, U4, and U6, and together with additional snRNAs and proteins, they form the **spliceosome**.



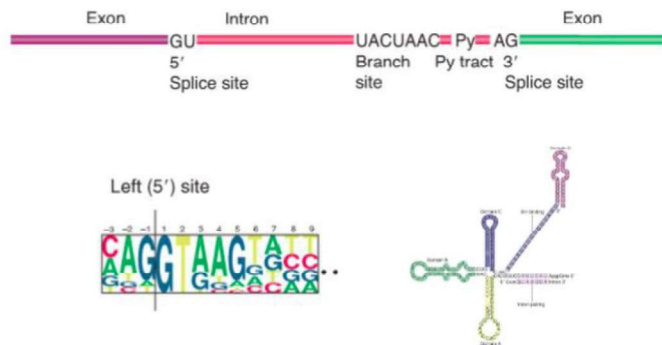
We need to know what the different snRNPs do

- U1

U1 is a complex of proteins and nucleotides

3. Splicing: the spliceosome – the process

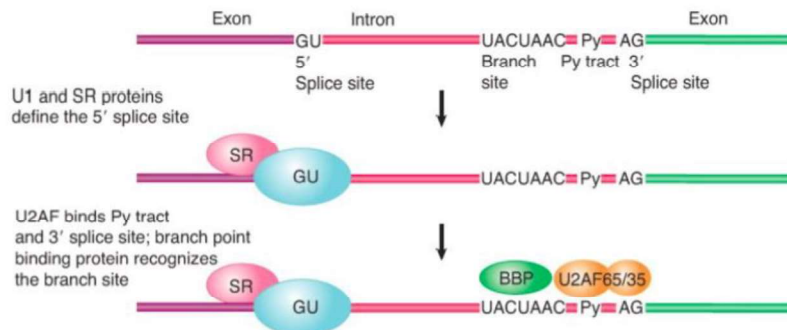
- U1 snRNP initiates splicing by binding to the 5' splice site by means of an RNA–RNA pairing reaction.



- U2AF

3. Splicing: the spliceosome – the process

- Next, the protein U2AF (or Mud2 in yeast) binds to the pyrimidine tract between the branch site and the 3' splice site, and the 65/35 protein and the Branch-Point-Protein (= SF1=BBP) is recruited to 3' splice site



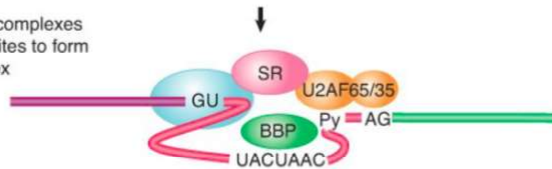
- Commitment complex – E complex

Early part of how introns will be spliced

3. Splicing: the spliceosome – the process

- The commitment complex (or **E complex – E for Early**) contains U1 snRNP bound at the 5' splice site and the protein U2AF, the 63/65 protein and the Branch-Point-Protein (=SF1) bound to the 3' splice site. The two splice sites are brought together by an SR protein

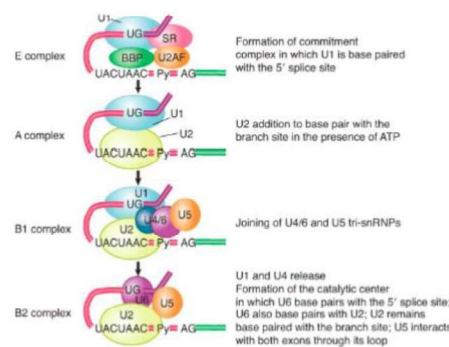
SR proteins bridge the complexes at the 5' and 3' splice sites to form commitment (E) complex



- The A, B1 and B2 complexes

3. Splicing: the spliceosome – the process

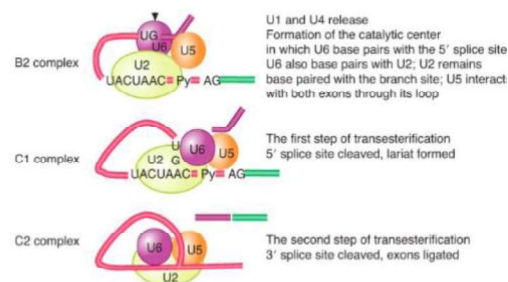
- The commitment complex progresses to prespliceosome (the **A complex**) in the presence of ATP.
- Recruitment of U5 and U4/U6 snRNPs converts the A complex to the mature spliceosome (the **B1 complex**).
- The B1 complex is next converted to the **B2 complex**, in which U1 and U4 snRNP are released to allow U6 snRNA to interact with the 5' splice site.



- B2, C1 and C2 complex

3. Splicing: the spliceosome – the process

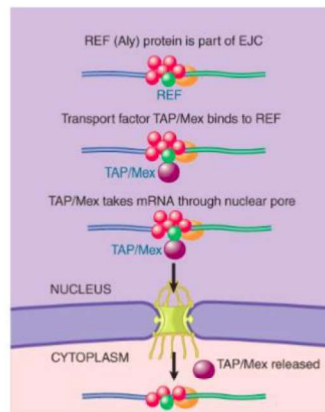
- When U4 dissociates from U6 snRNP, **U6 snRNA** can pair with **U2 snRNA** to form the **catalytic active site**.
- Both transesterification reactions take place in the activated spliceosome (the **C complex**).
- The spliced mRNA and the lariat is released from the complex
- The splicing reaction is reversible at all steps.



- The EJC (exon junction complex)

3. Splicing: the spliceosome – the process

- Once the introns are eliminated, the exons must be joined together.
- **Exon junction complex (EJC)** is a protein complex that assembles at exon-exon junctions during splicing
- It is also involved on assisting in RNA transport, localization, and degradation.



Every single step is reversible of all of these steps, if you have to, the most important thing to know is where the U2, U1 and U6 is placed and its function.

3.3 Autocatalytic introns

“The exception”

Some introns in prokaryotic cells that are self-spliced (?) meaning it doesn't go through the whole thing we just covered. But it folds into a new shape (adopt the shape of U6-U2 nuclear intron) which is responsible for packing the (?).

3.4 Alternative splicing/Trans-splicing

Fx something binds too good so the U1 cannot bind to 5' end anymore making it impossible for the cell to splice the intron (?). **what was it that made so that the U1 could not bind?**

Which statement regarding eukaryotic mRNA is TRUE?

mRNA is only matured by auto-splicing	##.##%
Translation is coupled to transcription	##.##%
Exons are excised from the pre-mRNA in the nucleus	##.##%
mRNA require modification prior to translocation to the cytoplasm	##.##%
	##.##%

Answer: mRNA require modification prior to translocation to the cytoplasm.

Which statement regarding splicing in eukaryotes is FALSE?

Catalytic RNA is part of the spliceosome

###.##%

Most splice sites can be recognised in the DNA sequence

###.##%

The spliceosome assemble at the pre-mRNA

###.##%

The branch site in RNA splicing is a protein sequence

###.##%

Answer: The branch site in RNA splicing is a protein sequence

What does the 5' cap of mRNA protect against

DNA polymerase attack

###.##%

Transcriptional silencing

###.##%

5' to 3' exonuclease degradation

###.##%

Histone methylation

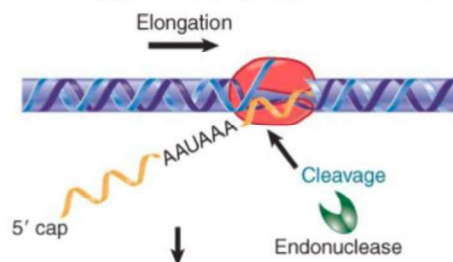
###.##%

Answer: 5' to 3' exonuclease degradation

4. Termination and polyadenylation

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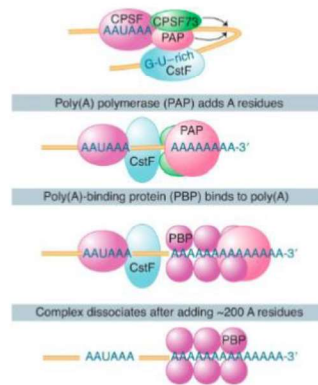
- In prokaryotes, termination happened due to the formation of a stem loop in the nascent mRNA which caused the disassembly of the RNA polymerase
- In eukaryotes the 3' ends of the mRNA are generated by cleavage and polyadenylation.
- This means that the polyA tail is not encoded in the DNA but added enzymatically to the mRNA after transcription
- The sequence AAUAAA is a signal for cleavage to generate a 3' end of mRNA



PAP – ploy(A) polymerase

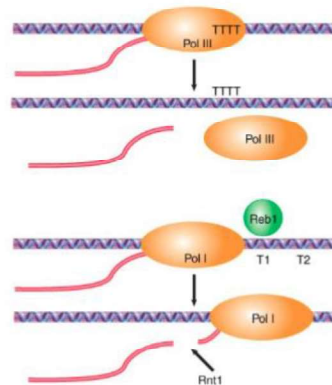
4. Termination and polyadenylation

- Cleavage Stimulatory Factor (CstF) binds to the GU-rich sequence downstream to the termination and recruits The cleavage and polyA specificity factors (CPSF) to the AAUAAA sequence
- CPSF cleaves the mRNA
- The formed 3' end of the RNA is substrate for the poly(A) polymerase (PAP)
- Poly(A) binding proteins (PBP = PABPII) controls length and ensures translocation to the cytoplasm.



4. Termination and polyadenylation

- Transcription can be ended in a number of different ways based on the type of RNA polymerase involved.
- Transcription by Pol I and Pol III uses specific terminators to end transcription.
- Pol I and III are not coupled with polyadenylation.



All of this is to protect the sides of mRNA which is the most likely to undergo degradation.

5. mRNA degradation

endonucleases “chews from the sides”

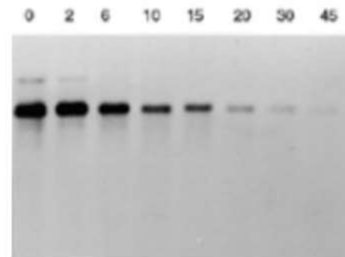
exonucleases attacks from the middle

they are very resistant – they are a pain

mRNA stability can be investigated through northern blot:

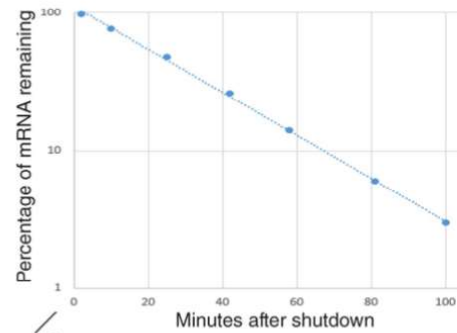
5. mRNA degradation

- mRNA stability can be investigated through Northern blot using a probe specific for the suspect mRNA
- The intensity of the bands can be densitometered and represented through the course of time.
- The half life is the amount of time necessary to decrease the concentration of mRNA 50% from the initial amount.
- The half life of mRNA varies greatly. Some are extremely stable (>1 hour) while others are gone in seconds.



5. mRNA degradation

- mRNAs are unstable molecules
- mRNA exhibit a wide range of half lives
- The half life of mRNA can be modified by
 - Introducing mutations that makes them more or less susceptible to RNases
 - Introducing mutations that makes them more or less susceptible to the binding of proteins that protects them or degrade them
 - Protecting or deprotecting the 5' and 3' edges



Shutdown of
RNA Pol II
transcription

Calculation of the half life is relevant here – how?

5.1 mRNA degradation: prokaryotes

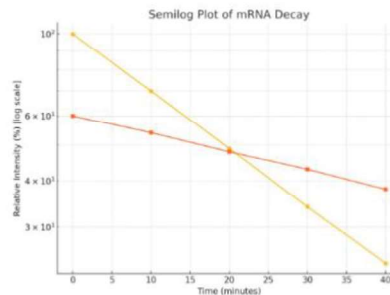
Kingdom	Enzyme	Specificity and/or function
Endonucleases		
Bacteria	RNase E* and RNase G*	Single-stranded RNA
	RNase III	Double-stranded RNA
	RNase J	Single-stranded RNA
	RNase Y	Single-stranded RNA
	Cmr complex	mRNA-CRISPR RNA duplexes
Eukaryotes	Argonaute	mRNA-sRNA or mRNA-mRNA duplexes that are fully paired
	SMG6	PTC-containing mRNAs
3' exonucleases		
Bacteria	Polynucleotide phosphorylase	Single-stranded 3' end
	RNase R	Single-stranded 3' end
	RNase II	Single-stranded 3' end
	Oligoribonuclease	RNA oligonucleotides
Eukaryotes	Exosome	3' end not protected by RNBP
5' exonucleases		
Bacteria	RNase J	Monophosphorylated 5' end
Eukaryotes	XRN1	Monophosphorylated 5' end

- Prokaryotes count with a rich variety of enzymes exclusively dedicated to degrade mRNA.
- It allows for a fast turn-around of resources and protects them against bacteriophages (virus) infections.

- investigating the half life of two mRNA's

Let's do it together

- You are investigating the half life of two mRNAs. You performed a northern blot and based on the intensity of the bands built the graph on the right representing the time (x-axis) vs the log₂ of the intensity (y-axis).
- Calculate the half-life of the two mRNAs
- Which mRNA has a longer half life, orange or yellow?**
 - Discuss how to extend the half life of mRNA



The half life of each:

From the graph (reading off approximate points):

- Yellow** line falls much more steeply; the distance for a 1-unit y-drop is about 8 min
 $\Rightarrow t_{1/2, \text{yellow}} \approx 8 \text{ min.}$
- Orange** line is shallower; a 1-unit y-drop takes about 20 min
 $\Rightarrow t_{1/2, \text{orange}} \approx 20 \text{ min.}$

So the **orange** mRNA has the longer half-life (~20 min vs ~8 min).

The mRNA with the longer half life:

Steeper slope → faster decay → shorter half-life

Shallower slope → slower decay → longer half-life

- The **yellow line** drops steeply → very short half-life.
- The **orange line** decreases more gradually → longer half-life.

So The orange mRNA has a longer half-life than the yellow mRNA.

How to extend the half life of mRNA:

Cells (and scientists) can extend mRNA stability by:

- Adding or protecting the 5' cap.
- Lengthening the poly-A tail.
- Using RNA-binding proteins that stabilize the 3' UTR.
- Modifying UTR sequences to reduce recognition by decay enzymes.
- Avoiding destabilizing microRNA binding sites.

Which of the following best describes a lariat structure in splicing?

Double-stranded RNA segment	##.##%
Loop formed by joining the 5' end of intron to a branch point A	##.##%
Linear mRNA with capped ends	##.##%
Reverse intron-exon order	##.##%
	##.##%

Answer: loop formed by joining the 5' end of intron to a branch point A.

What is the purpose of deadenylation in mRNA decay?

Enhance translation	##.##%
Initiate degradation by exonucleases	##.##%
Increase the stability	##.##%
Cap the 5' end	##.##%
	##.##%

Answer: initiate degradation by exonucleases

5.1 Prokaryotes

5.2 Eukaryote

Molecular Biology Week 4: Translation

Learning objectives

- Explain in your own words the composition of ribosomes, and their subunits, and active sites, A, P and E site
- Explain the function of tRNA
- Describe the different stages in translation: Initiation, elongation and termination
- Describe the general mechanism in prokaryotic translation initiation
- Identify the sequence of the ribosome binding site (Shine-Dalgarno sequence)
- Explain why all proteins starts with a formyl-methionine in prokaryotes
- Describe the general mechanism in eukaryotic translation initiation
- Present the name and the role of the initiation and elongation co-factors in the initiation and elongation process
- Explain how the ribosome/mRNA moves during the translation process
- Define the termination of translation and what co-factors are involved
- Describe the concept of the genetic code and how several codons may code for the same amino acid and what is meant by wobble recognition
- Explain the consequences of codon usage in protein translation